

Independent relative *whose*

Fragment 2 — 2026-05-07

This fragment tests whether Pullum’s bridge-linked dominance-chain machinery can be used for an empirically delicate CGEL construction: relative *whose* without an overt following possessum.

1. Phenomenon

Independent relative *whose* is the relative-genitive construction in which *whose* appears without a following noun:

- (1) *I was going to visit Lucy, a friend of whose had told us of the accident.*
- (2) *Norway, whose standard of living is the envy of the world, has thicker kids than Singapore, whose isn’t.*
- (3) *The man whose these are hath gotten me with child.*

The construction is rare and pragmatically constrained. The syntactic task for this fragment is narrower: state the tree condition under which a relative clause contains a relative-genitive phrase whose possessum is not overtly realised, and connect that phrase to a vacancy or predicative gap in the relative-clause nucleus.

2. Vocabulary

The definitions below adapt Pullum’s handout notation. Immediate dominance is $\triangleleft$; dominance is $\triangleleft^*$. The relation $\text{Bridge}(x, y)$ is initially limited to Head and Complement links:

- (4) $\text{Bridge}(x, y) \equiv_{\text{def}} \text{Head}(x, y) \vee \text{Comp}(x, y)$
y is linked to x by a bridge-preserving function

We add $\text{RelWhose}(w)$ for a lexical relative-genitive *whose* node and $\text{PossessumRecoverable}(w)$ for the discourse condition that the missing possessum is recoverable. The second predicate is not syntactic; it marks the point where this construction is known to need information-structural licensing.

3. Specification

Pullum’s basic vacancy predicate can be stated as:

- (5) $\text{Vacancy}(x) \equiv_{\text{def}} \text{Phrasal}(x) \wedge \neg(\exists y)[x \triangleleft y]$
a phrasal node with no children
- (6) $\text{NPVacancy}(x) \equiv_{\text{def}} \text{Vacancy}(x) \wedge \text{NP}(x)$
an NP vacancy

The relevant chain definitions are:

- (7) $\text{DomOrdered}(X) \equiv_{\text{def}} (\forall x)(\forall y)[(X(x) \wedge X(y)) \Rightarrow (x \triangleleft^* y \vee y \triangleleft^* x)]$.
all nodes in X are ordered by dominance
- (8) $\text{DomContinuous}(X) \equiv_{\text{def}} (\forall x)(\forall y)(\forall z)[(X(x) \wedge X(z) \wedge x \triangleleft^* y \wedge y \triangleleft^* z) \Rightarrow X(y)]$.
X contains the intervening nodes on its dominance path
- (9) $\text{DomChain}(X) \equiv_{\text{def}} \text{DomOrdered}(X) \wedge \text{DomContinuous}(X)$
a dominance chain
- (10) $\text{BridgeLinked}(X) \equiv_{\text{def}} \text{DomChain}(X) \wedge (\forall x)(\forall y)[(X(x) \wedge X(y) \wedge x \triangleleft y) \Rightarrow \text{Bridge}(x, y)]$.
a dominance chain whose immediate links are bridge links
- (11) $\text{Root}(X, r) \equiv_{\text{def}} (\forall x)[X(x) \Leftrightarrow r \triangleleft^* x]$
X is the set of nodes dominated by root r
- (12) $\text{NPBindable}(r) \equiv_{\text{def}} (\exists X)[\text{Root}(X, r) \wedge \text{BridgeLinked}(X) \wedge (\exists z)[X(z) \wedge \text{NPVacancy}(z)]]$.
r roots a bridge-linked constituent containing an NP vacancy

For independent relative *whose*, the draft adds a condition on the relative clause:

- (13) $\text{IndRelWhoseClause}(c) \equiv_{\text{def}} \text{Clause}(c) \wedge (\exists w)(\exists r)[\text{RelWhose}(w) \wedge c \triangleleft^* w \wedge c \triangleleft^* r$
 $\wedge \text{NPBindable}(r) \wedge \text{PossessumRecoverable}(w)]$.
a relative clause containing independent whose and a bindable nucleus

4. Worked instance

In *Singapore, whose isn't*, the relative clause contains a relative-genitive *whose* and a nucleus with stranded auxiliary *isn't*. The missing possessum is recoverable from the parallel antecedent *Norway, whose standard of living is* In this use, $\text{PossessumRecoverable}(w)$ is licensed by contrastive parallelism.

The syntactic part then asks whether the nucleus rooted at r contains a bridge-linked NP vacancy. If it does, r satisfies $\text{NPBindable}(r)$ and the relative clause satisfies $\text{IndRelWhoseClause}(c)$. The formula therefore keeps two claims apart: the structural possibility of a relative-genitive dependency, and the pragmatic recoverability of the possessum.

5. Open questions

- (a) Some attested cases are predicative-complement cases rather than simple NP-vacancy cases. Those may require a second vacancy type.
- (b) Oblique genitive examples such as *a friend of whose* may be locally licensed by the possessum noun inside the relative phrase, rather than by a distant vacancy in the nucleus.
- (c) $\text{PossessumRecoverable}$ is a placeholder for discourse licensing, especially contrastive parallelism, deictic anchoring, and structural anchoring. The syntax fragment should say where that condition attaches, but not pretend that it is a tree-only predicate.
- (d) CGEL page references are available in the separate *whose* project but should be rechecked against the book before citation.